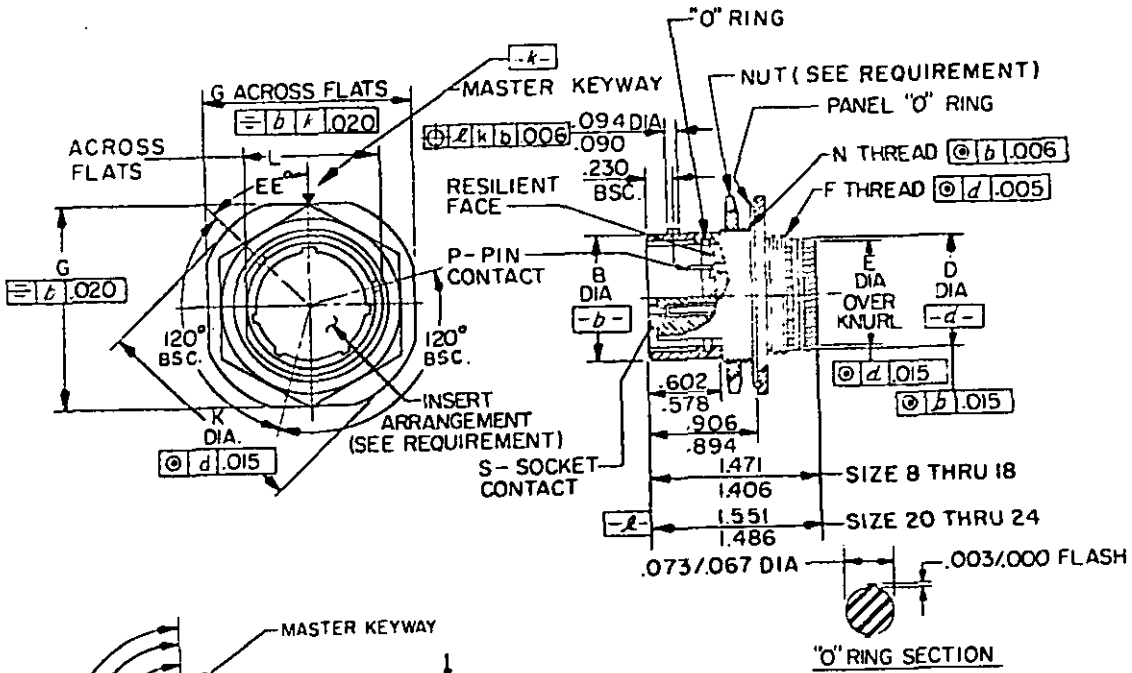
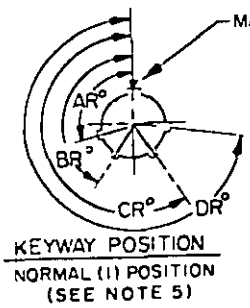
MILITARY SPECIFICATION SHEET

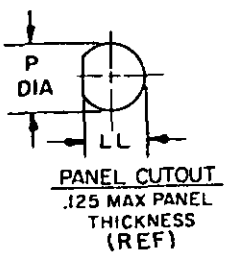
# CONNECTORS, ELECTRICAL, CIRCULAR, HIGH DENSITY, QUICK DISCONNECT, ENVIRONMENT RESISTING: AND ACCESSORIES: RECEPTACLE, JAM NUT MOUNT, CRIMP-TYPE CONTACTS, CLASS A, F AND E (SERIES 1)

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-81511.

This specification is mandatory for use by all Department and Agencies of the Department of Defense.




| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .003 | .08 | .230 | 5.84 |
| .005 | .13 | .578 | 14.68 |
| .006 | .15 | .602 | 15.29 |
| .015 | .38 | .894 | 22.71 |
| .020 | .51 | .906 | 23.01 |
| .067 | 1.70 | 1.406 | 35.71 |
| .073 | 1.85 | 1.436 | 36.47 |
| .090 | 2.29 | 1.471 | 37.36 |
| .094 | 2.39 | 1.531 | 38.89 |
| .125 | 3.18 | | |



FSC 5935

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TABLE I.

| Shell size | B dia | D dia | E dia | F thread | G across flat | K dia | L across flat | N thread | P dia | LL |
|---|---|---|---|---|---|---|---|---|---|---|
| 8 | .553 (14.05) .547 (13.89) | .430 (10.92) .424 (10.77) | .422 (10.72) .407 (10.34) | ½-28UNEF-2A | .989 (25.12) .969 (24.61) | 1.072 (27.23) 1.052 (26.72) | .598 (15.19) .590 (14.99) | 5/8-20UN-2A | .640 (16.26) .630 (16.00) | .610 (15.49) .600 (15.24) |
| 10 | .678 (17.22) .672 (17.07) | .555 (14.10) .549 (13.94) | .548 (13.92) .533 (13.54) | 5/8-28UN-2A | 1.114 (28.30) 1.094 (27.79) | 1.197 (30.40) 1.177 (29.90) | .723 (18.36) .715 (18.16) | 3/4-20UNEF-2A | .765 (19.43) .755 (19.18) | .735 (18.67) .725 (18.42) |
| 14 | .928 (23.57) .922 (23.42) | .805 (20.45) .799 (20.29) | .798 (20.27) .783 (19.89) | 7/8-28UN-2A | 1.401 (35.59) 1.381 (35.08) | 1.510 (38.35) 1.490 (37.85) | .973 (24.71) .965 (24.51) | 1-20 UNEF-2A | 1.015 (25.78) 1.005 (25.53) | .985 (25.02) .975 (24.77) |
| 16 | 1.054 (26.77) 1.048 (26.62) | .930 (23.62) .924 (23.47) | .924 (23.47) .909 (23.09) | 1-28UN-2A | 1.526 (38.76) 1.506 (38.25) | 1.635 (41.53) 1.615 (41.02) | 1.098 (27.89) 1.090 (27.69) | 1-1/8-20UN-2A | 1.140 (26.42) 1.130 (28.70) | 1.110 (28.19) 1.100 (27.94) |
| 18 | 1.178 (29.92) 1.172 (29.77) | 1.055 (26.80) 1.049 (26.64) | 1.048 (26.62) 1.033 (26.24) | 1-1/8-28UN-2A | 1.651 (41.94) 1.631 (41.43) | 1.745 (44.32) 1.725 (43.82) | 1.222 (31.04) 1.214 (30.84) | 1-1/4-20UN-2A | 1.265 (32.13) 1.255 (31.88) | 1.235 (31.37) 1.225 (31.12) |
| 20 | 1.303 (33.10) 1.297 (32.94) | 1.180 (29.97) 1.174 (29.82) | 1.173 (29.79) 1.158 (29.41) | 1¼-28UN-2A | 1.776 (45.11) 1.756 (44.60) | 1.870 (47.50) 1.850 (46.99) | 1.347 (34.21) 1.339 (34.01) | 1 3/8-20UN-2A | 1.390 (35.31) 1.380 (35.05) | 1.360 34.54) 1.350 (34.29) |
| 22 | 1.428 (36.27) 1.422 (36.12) | 1.305 (33.15) 1.299 (32.99) | 1.298 (32.97) 1.283 (32.59) | 1 3/8-28UN-2A | 1.901 (48.29) 1.881 (47.78) | 1.995 (50.67) 1.975 (50.17) | 1.472 (37.39) 1.464 (37.19) | 1½-20UN-2A | 1.515 (38.48) 1.505 (38.23) | 1.485 (37.72) 1.475 (37.47) |
| 24 | 1.553 (39.45) 1.547 (39.29) | 1.430 (36.32) 1.424 (36.17) | 1.423 (36.14) 1.408 (35.76) | 1½-28UN-2A | 2.026 (51.46) 2.006 (50.95) | 2.120 (53.85) 2.100 (53.34) | 1.597 (40.56) 1.589 (40.36) | 1 5/8-20UN-2A | 1.640 (41.66) 1.630 (41.40) | 1.610 (40.89) 1.600 (40.64) |



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TABLE I. - Continued

| Shell size | AR• BSC | BR• BSC | CR• BSC | DR• BSC | Panel "O" ring | EE• BSC Position | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 1 | 2 | 3 | 4 | 5 | 6 |
| 8 | 105 | 140 | 215 | 265 | MS9068-018 | 60 | 46 | 43 | 60 | 89 | 43 |
| 10 | 95 | 141 | 208 | 236 | MS9068-020 | 53 | 87 | 51 | 77 | 87 | 87 |
| 14 | 95 | 141 | 208 | 236 | MS9068-024 | 53 | 87 | 51 | 77 | 87 | 87 |
| 16 | 80 | 142 | 196 | 293 | MS9068-026 | 98 | 98 | 98 | 98 | 98 | 98 |
| 18 | 80 | 142 | 196 | 293 | MS9068-028 | 98 | 98 | 98 | 98 | 98 | 98 |
| 20 | 80 | 142 | 196 | 293 | MS9068-029 | 98 | 98 | 98 | 98 | 98 | 98 |
| 22 | 80 | 142 | 196 | 293 | MS9068-030 | 98 | 98 | 98 | 98 | 98 | 98 |
| 24 | 80 | 142 | 196 | 293 | MS9068-031 | 98 | 98 | 98 | 98 | 98 | 98 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions are after plating.
5. Normal mating keyway position shown. For alternate keyway positions see MIL-C-81511, Figure 8.



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REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Series: 1

Classes: A, F and E.

Shell size: See table II.

Insert arrangement: See supplement for MS sheet corresponding to desired shell size.

Contacts: Style P or S, crimp type.

Shell polarization: 1 thru 6, key position.

Weight: See table II.

Marking: Part number on shell.

Mating counterpart: Plug connectors specified in MIL-C-81511/26 and /38.

Insertion and removal tool: As specified in MS3323, MS3342 and MS3344.

Slotted nut: Shall conform to MIL-C-81511/20.

Part number: The part number shall consist of M81511/23 together with the applicable codes for class, shell size, insert arrangement, contact style, and key positions. Information concerning the formulation of the part number can be found in MIL-C-81511.

TABLE II.

| Shell size | Weight (REF) 1/ (pounds) |
|---|---|
| 24 (code J) | .151 |
| 22 (code H) | .137 |
| 20 (code G) | .119 |
| 18 (code F) | .100 |
| 16 (code E) | .082 |
| 14 (code D) | .070 |
| 10 (code B) | .046 |
| 8 (code A) | .035 |

1/ Removable crimp contacts not included in weight.
(See contact drawings for weight of pin and socket contact.)

Custodians:
Army - EL
Navy - AS
Air Force - 17

Preparing activity:
Navy - AS

Review activities:
Army - MI, MU, AV, WC
Navy - SH, EC, OS
Air Force - 11, 80
DSA - ES

(Project No. 5935-1732-17)

Uaer activities:
Army - ME, AT
Navy - MC
Air Force -



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